HW2.1

SCIE 001 Math

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1. Suppose that f is differentiable with $f'(x) > 0$ for all x and suppose that $f(1) = 0$. Set

$$F(x) = \int_0^x f(t)\,\mathrm{d}t$$

Determine if each statement is True or False. Justify your answer.

(a) The function $F(x)$ is continuous.

For $F(x)$ to be continuous for all x:

$$\lim_{a \to x} F(a) = F(x)$$

Therefore we must prove:

$$\lim_{a \to x} \int_0^a f(t)\,\mathrm{d}t = \int_0^x f(t)\,\mathrm{d}t$$

To do this, we can show that we can rearrange the left side to equal the right side.

Expanding the integral gives:

$$\lim_{a \to x} \left(\lim_{n \to \infty} \sum_{i=1}^{n} \frac{a-0}{n} f\left(\frac{i(a-0)}{n} + 0 \right) \right)$$
$$= \lim_{a \to x} \left(\lim_{n \to \infty} \sum_{i=1}^{n} \frac{a}{n} f\left(i\frac{a}{n} \right) \right)$$

Since we know $f(x)$ is differentiable, it must also be continuous, as differentiability implies continuity. For any $n \in \mathbb{R}, n \neq 0$, $\frac{a}{n}$ is continuous for all a. Consequently, for any $i \in \mathbb{R}$:

$$\frac{a}{n} f\left(i\frac{a}{n} \right)$$

is also continuous for all a.

Finally, the sum of any continuous functions is also continuous. Therefore:

$$\sum_{i=1}^{n} \frac{a}{n} f\left(i\frac{a}{n} \right)$$

is continuous for all a.

So, we can rewrite our limit:

$$= \lim_{a \to x} \left(\lim_{n \to \infty} \sum_{i=1}^{n} \frac{a}{n} f\left(i\frac{a}{n} \right) \right)$$
$$= \lim_{n \to \infty} \sum_{i=1}^{n} \frac{x}{n} f\left(i\frac{x}{n} \right)$$

(b) The function $F(x)$ is not twice differentiable.

False

$$\frac{\mathrm{d}}{\mathrm{d}x}F(x) = \frac{\mathrm{d}}{\mathrm{d}x}\int_0^x f(t)\,\mathrm{d}t$$

By the Fundamental Theorem of Calculus:

$$\frac{\mathrm{d}}{\mathrm{d}x}F(x) = f(x)$$

Taking a second derivative:

$$\frac{\mathrm{d}^2}{\mathrm{d}^2x}F(x) = \frac{\mathrm{d}}{\mathrm{d}x}f(x)$$

By definition, f is differentiable for all x, hence $\frac{\mathrm{d}}{\mathrm{d}x}f(x)$ exists for all x, therefore $\frac{\mathrm{d}^2}{\mathrm{d}^2x}F(x)$ exists for all x.

Therefore $F(x)$ is twice differentiable.

(c)
> The value x = 1 is a critical point of F.

True

For $x = 1$ to be a critical point of F:

$$\frac{\mathrm{d}}{\mathrm{d}x}F(1) = 0$$

From part b we know:

$$\frac{\mathrm{d}}{\mathrm{d}x}F(x) = f(x)$$

and by definition we know:

$$f(1) = 0.$$

Hence:

$$\frac{\mathrm{d}}{\mathrm{d}x}F(1) = f(1) = 0$$

Therefore, $F(x)$ has a critical point at $x = 1$.

(d)
> The function $F(x)$ takes on a local minimum at $x = 1$.

(e)
> $F(1) > 0$.

> Based on your answers above, make a rough sketch of the graph of $F(x)$.

2.
> Recall in class we proved that $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$. In this problem, we will construct a proof of
>
> $S = \sum_{i=1}^{k} i^2$

(a) Calculate the value of S for $k = 1, 2, 3$ and 4.

$$S_{k=1} = \sum_{i=1}^{1} i^2 = 1^2 = 1$$

$$S_{k=2} = \sum_{i=1}^{2} i^2 = 1^2 + 2^2 = 3$$

$$S_{k=3} = \sum_{i=1}^{3} i^2 = 1^2 + 2^2 + 3^2 = 14$$

$$S_{k=4} = \sum_{i=1}^{4} i^2 = 1^2 + 2^2 + 3^2 + 4^2 = 30$$

(b) Calculate the value of $\dfrac{k(k+1)(2k+1)}{6}$ for $k = 1, 2, 3,$ and 4

$$k = 1 \qquad \frac{1(1+1)(2(1)+1)}{6} = 1$$

$$k = 2 \qquad \frac{2(2+1)(2(2)+1)}{6} = 5$$

$$k = 3 \qquad \frac{3(3+1)(2(3)+1)}{6} = 14$$

$$k = 4 \qquad \frac{4(4+1)(2(4)+1)}{6} = 30$$

(c) Show that $\sum_{i=1}^{k} i^2 = \dfrac{k(k+1)(2k+1)}{6}$ by constructing a proof by induction by:

i. showing the equality is true for the base case (i.e., $k = 1$).

ii. for each positive integer N, the equality holds for $k = N$ implies the equality for $k = N + 1$.

Be sure to end your proof with an appropriate conclusion statement (i.e., "Therefore, by induction, …").

Lemma 1 : $\boldsymbol{\sum_{i=1}^{k} i^2 = \dfrac{k(k+1)(2k+1)}{6}}$ **holds for** $\boldsymbol{k = 1}$.

Left Hand	Right hand
$\sum_{i=1}^{1} i^2 = 1^2 = 1$	$\dfrac{1(1+1)(2(1)+1)}{6} = 1$

$$\therefore \sum_{i=1}^{1} i^2 = \frac{1(1+1)(2(1)+1)}{6}$$

Therefore $\sum_{i=1}^{k} i^2 = \dfrac{k(k+1)(2k+1)}{6}$ holds for $k = 1$.

Lemma 2 : For each positive integer N, the equality holding for $k = N$ implies the equality for $k = N + 1$.

Let $f(x) = \sum_{i=1}^{x} i^2$. Let $g(x) = \frac{x(x+1)(2x+1)}{6}$

Let n be some positive integer such that the equality $\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}$ holds.

Therefore $f(n) = g(n)$

Consider $f(n+1)$.

$$f(n+1) = \sum_{i=1}^{n+1} i^2 = \left(\sum_{i=1}^{n} i^2\right) + (n+1)^2$$
$$f(n+1) = f(n) + (n+1)^2$$

Then consider $g(n+1)$.

$$\begin{aligned} g(n+1) &= \frac{(n+1)((n+1)+1)(2(n+1)+1)}{6} \\ &= \frac{(n+1)(n+2)(2n+3)}{6} \\ &= \frac{(2n^3+9n^2+13n+6)}{6} \\ &= \frac{(2n^3+3n^2+1n)+(6n^2+12n+6)}{6} \\ &= \frac{n(2n^2+3n+1)}{6} + (n^2+2n+1) \\ &= \frac{n(2n+1)(n+1)}{6} + (n+1)^2 \\ &= g(n) + (n+1)^2 \end{aligned}$$

Putting them together, by definition

$$f(n) = g(n)$$
$$f(n) + (n+1)^2 = g(n) + (n+1)^2$$
$$\therefore f(n+1) = g(n+1)$$

Therefore for every $N \in \mathbb{Z}^+$, $f(N) = g(N)$ implies that $f(N+1) = g(N+1)$. i.e the equality holding for $k = N$ implies that it holds for $k = N + 1$

Therefore by induction, since $\sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}$ holds for $k = 1$ and this equality holding for $k = N$ implies that it holds for $k = N + 1$ provided that $N \in \mathbb{Z}^+$:

$$S = \sum_{i=1}^{k} i^2 = \frac{k(k+1)(2k+1)}{6}$$

holds for all $k \in \mathbb{Z}^+$

3. Let $f(x)$ be an integrable function over the interval $[a, b]$. Choose an integer N and let

$$x_k = a + k\Delta x, \quad k = 0, 1, 2, ..., N \text{ where } \Delta x = \frac{b-a}{N}$$

The trapezoid rule is an approximation of the definite integral:

$$\int_a^b f(x)\,\mathrm{d}x \approx \Delta x \sum_{k=1}^{N} \frac{f(x_k) + f(x_{k-1})}{2},$$

where $\Delta x = \frac{b-a}{N}$

A bound on the error is given by the error formula

$$\left| \int_a^b f(x)\,\mathrm{d}x - \frac{b-a}{N} \sum_{k=1}^{N} \frac{f(x_k) + f(x_{k-1})}{2} \right| \leq \frac{(b-a)^3}{12N^2} K_2$$

where K_2 is any number such that $|f''(x)| \leq K_2$ for all $x \in [a, b]$.

Consider the Fresnel integral

$$\int_0^{\sqrt{\frac{\pi}{2}}} \sin(x^2)\,\mathrm{d}x$$

(a) Write out and calculate the Left and Right Riemann sums for $N = 4$.

(b) Write our and calculate the Trapezoid rule for $N = 4$.

(c) Find and appropriate value of K_2 for the Fresnel integral. Be sure to justify your answer.

(d) Find the value of (the smallest) N which guarantees the trapezoid rule approximates the Fresnel Integral with errors less than 10^{-1} and 10^{-2}. Be sure to justify your answers.

4. In class, we proved Part 2 of the Fundamental Theorem of Calculus, which states that if f is a continuous function on the interval $[a, b]$ and F is any antiderivative of f, then

$$F(b) - F(a) = \int_a^b f(x)\,\mathrm{d}x$$

Here, you will construct another proof of Part 2 of the Theorem.

(a) Divide the interval $[a, b]$ in n subintervals with endpoints $a = x_0 < x_1 < x_2 < ... < x_n = b$. Show that

$$F(b) - F(a) = \sum_{i=1}^{n} (F(x_i) - F(x_{i-1}))$$

Let $a = x_0 < x_1 < x_2 < ... < x_n = b$

Consider the sum:

$$\sum_{i=1}^{n} (F(x_i) - F(x_{i-1}))$$

Expanding, we get:

$$\sum_{i=1}^{n} (F(x_i) - F(x_{i-1}))$$
$$= \left(\sum_{i=1}^{n} F(x_i) \right) - \sum_{i=1}^{n} F(x_{i-1}))$$
$$= (F(x_1) + F(x_2) + ... + F(n-1) + F(x_n)) - (F(x_0) + F(x_1) + F(x_2) + ... + F(x_n - 1))$$

All terms except for $F(x_0)$ and $F(x_n)$ cancel, leaving:

$$\sum_{i=1}^{n} (F(x_i) - F(x_{i-1})) = F(x_n) - F(x_0)$$

By definition $x_n = b$ and $x_0 = a$, therefore

$$\sum_{i=1}^{n} (F(x_i) - F(x_{i-1})) = F(b) - F(a)$$

(b) Now suppose that F is any antiderivative of f. Show that there exists a number c_i in each interval $[x_{i-1}, x_i]$ such that

$$F(x_i) - F(x_{i-1}) = f(c_i)(x_i - x_{i-1})$$

(c) Now build an appropriate Riemann sum for f on $[a, b]$ and show that

$$F(b) - F(a) = \int_a^b f(x) \, \mathrm{d}x$$

A Guide to Finding Intersection Points of P_{in} and P_{out} in Python

Create a new Jupyter notebook using anaconda. Copy and paste the following lines of code into a new cell.

1. **<u>Your task:</u>** Derive the EBM given above and verify that the units on both sides of the

Left Side	**Right Side**
$= JK^{-1}\frac{K}{s}$ $= \frac{J}{s}$ $= W$	$= W - W$ $= W$

$$W = W$$

Left Side = Right Side

Therefore, the units are consistent.

$$\lim_{T\to 247^-} \alpha(T) = \alpha(247) \qquad \lim_{T\to 247^+} \alpha(T) = \alpha(247) \qquad \lim_{T\to 282^-} \alpha(T) = \alpha(282) \qquad \lim_{T\to 282^+} \alpha(T) = \alpha(282)$$